

Week 3 - ASP.NET Core Web API

Handson 4 - Web API CRUD Operation

Objective

- Perform CRUD operations using ASP.NET Core Web API.
- Update employee information using HTTP PUT.
- UseFromBody attribute.
- Validate employee ID.
- Test using Swagger.

Software Requirements

.NET 8 SDK

Visual Studio Code / Visual Studio 2022

ASP.NET Core Web API

Swagger

Folder Structure

```
WebApi_Handson4/  
├── WebApi_Handson4.csproj  
├── Program.cs  
├── appsettings.json  
├── appsettings.Development.json  
├── Controllers/  
│   └── EmployeeController.cs  
├── Models/  
│   ├── Employee.cs  
│   ├── Department.cs  
│   └── Skill.cs  
├── Properties/  
│   └── launchSettings.json  
└── README.md
```

WebApi_Handson4.csproj

```
<Project Sdk="Microsoft.NET.Sdk.Web">  
  <PropertyGroup>  
    <TargetFramework>net8.0</TargetFramework>  
    <Nullable>enable</Nullable>  
    <ImplicitUsings>enable</ImplicitUsings>  
  </PropertyGroup>  
  <ItemGroup>  
    <PackageReference Include="Swashbuckle.AspNetCore" Version="6.6.2"/>  
  </ItemGroup>  
</Project>
```

Program.cs

```
var builder=WebApplication.CreateBuilder(args);
builder.Services.AddControllers();
builder.Services.AddEndpointsApiExplorer();
builder.Services.AddSwaggerGen();
var app=builder.Build();
app.UseSwagger();
app.UseSwaggerUI();
app.UseHttpsRedirection();
app.UseAuthorization();
app.MapControllers();
app.Run();
```

Department.cs

```
namespace WebApi_Handson4.Models{ public class Department{ public int Id{get;set;}
public string Name{get;set;}=string.Empty; } }
```

Skill.cs

```
namespace WebApi_Handson4.Models{ public class Skill{ public int Id{get;set;} public
string Name{get;set;}=string.Empty; } }
```

Employee.cs

```
namespace WebApi_Handson4.Models{ public class Employee{ public int Id{get;set;} public
string Name{get;set;}=string.Empty; public int Salary{get;set;} public bool
Permanent{get;set;} public Department Department{get;set;}=new(); public List<Skill>
Skills{get;set;}=new(); public DateTime DateOfBirth{get;set;} } }
```

EmployeeController.cs

```
[ApiController]
[Route("api/[controller]")]
public class EmployeeController:ControllerBase{
    private static List<Employee> employees=new();
    [HttpGet] public ActionResult<List<Employee>> Get()=>Ok(employees);
    [HttpPut("{id}")] public ActionResult<Employee> Put(int id,[FromBody] Employee
employee){ if(id<=0) return BadRequest("Invalid employee id"); var
emp=employees.FirstOrDefault(e=>e.Id==id); if(emp==null) return BadRequest("Invalid
employee id"); emp.Name=employee.Name; emp.Salary=employee.Salary;
emp.Permanent=employee.Permanent; emp.Department=employee.Department;
emp.Skills=employee.Skills; emp.DateOfBirth=employee.DateOfBirth; return Ok(emp);} }
```

appsettings.json

```
{
  "Logging":{"LogLevel":{"Default":"Information"}},
  "AllowedHosts": "*"
}
```

appsettings.Development.json

```
{
  "Logging":{"LogLevel":{"Default":"Debug"}}
}
```

launchSettings.json

```
{ "profiles":{ "http":{ "launchUrl": "swagger", "applicationUrl": "http://localhost:5200" } } }
```

README.md

Run:

```
dotnet restore
```

```
dotnet build
```

```
dotnet run
```

Open <http://localhost:5200/swagger>

Testing

Run:

```
dotnet restore
```

```
dotnet build
```

```
dotnet run
```

Open <http://localhost:5200/swagger>

Test GET `/api/Employee` and PUT `/api/Employee/{id}`

Conclusion

This hands-on demonstrates implementing an HTTP PUT API to update employee information with validation and Swagger testing.